

# *“The Essentials of Computer Organization and Architecture”*

## Unit 7 – Exercises and Solutions

FM1 On computer X, running a certain benchmark took 50 minutes. This computer is 13% faster than computer Y running the same benchmark. Determine:

- a) The time it takes for computer Y to run the benchmark.
- b) The acceleration of computer Y in relation to computer X.

$$S(A,B) = T(B)/T(A)$$

= *speedup* of A relative to B

$$[ S(A,B) - 1 ] \times 100 = m\%$$

FM1 On computer X, running a certain benchmark took 50 minutes. This computer is 13% faster than computer Y running the same benchmark. Determine:

**a) The time it takes for computer Y to run the benchmark.**

**Resolution:**

- according with the problem statement  
 $T(X) = 50 \text{ m} = 3000\text{s}$  and  $S(X,Y) = 1.13$

- once  $S(X,Y) = T(Y) / T(X)$ ,

then  $1.13 = T(Y) / 3000$

and thus  $T(Y) = 1.13 * 3000 = 3390 \text{ s}$

FM1 On computer X, running a certain benchmark took 50 minutes. This computer is 13% faster than computer Y running the same benchmark. Determine:

**b) The acceleration of computer Y in relation to computer X.**

**Resolution:**

- it is known that  $T(X) = 3000s$  and in a) it was calculated  $T(Y) = 3390s$
- by definition,  $S(Y,X) = T(X) / T(Y) = 3000 / 3390 = \mathbf{0.8849}$
- thus, Y has 88.40% of X's performance
- or, equivalently, Y has 11.51% less performance than X  
(  $(1-0.8849) \times 100 = 11.51\%$  )

# FM2 Pay attention to the scenario described below, and present what is requested.

In a software development company, there are 8 working hours per day. During these 8 hours, computers are used for code editing (task x) 66% of the time, and for compilation (task y) 33% of the time. Outside of working hours, computers are kept turned on to train neural networks (task z).

Before buying new computers for the company, suppliers are asked to run a benchmark with 3 micro-benchmarks:  $B_x$  (evaluates the capacity to edit code),  $B_y$  (measures the capacity to compile code) and  $B_z$  (measures the capacity to train neural networks). The result of each micro-benchmark is  $R_x$ ,  $R_y$  and  $R_z$ .

- a) Present an adequate metric of the average performance of a computer when executing tasks x, y and z in such company.
- b) Present an adequate metric to compare the performance of two different types of computers, a and b, when executing tasks x, y and z.
- c) If  $C_a$  and  $C_b$  are the cost of computer types a and b, present an adequate metric to select the most economically attractive computer.

# FM2 Pay attention to the scenario described below, and present what is requested.

## Resolution:

a) Present an adequate metric of the average performance of a computer when executing tasks x, y and z in such company.

- the day has 24 hours
- the working hours are 8 hours:  $8/24 = 1/3 = 33\%$  of the day
- there are 16 hours outside the working hours:  $16/24 = 2/3 = 66\%$  of the day
- the fraction of the day used to edit code is  $33\% * 66\% = 21.78\%$
- the fraction of the day used to compile code is  $33\% * 33\% = 10.89\%$
- the fraction of the day used to train neural networks is  $66\%$
- average performance can be measured via the **weighted average**

$$P = 0.2178 * R_x + 0.1089 * R_y + 0.66 * R_z$$

FM2 Pay attention to the scenario described below, and present what is requested.

### **Resolution (cont.):**

**b) Present an adequate metric to compare the performance of two different types of computers, a and b, when executing tasks x, y and z.**

- let  $P_a$  and  $P_b$  be the average performance of computer types a and b
- the speedup metric, based on the weighted averages  $P_a$  and  $P_b$ , is an adequate metric to evaluate the relative performance of types a and b

$$S(a,b) = P_a / P_b$$

$$S(b,a) = P_b / P_a$$

FM2 Pay attention to the scenario described below, and present what is requested.

### Resolution (cont.):

c) If  $C_a$  and  $C_b$  are the cost of computer types a and b, present an adequate metric to select the most economically attractive computer.

- for each type a and b, the average performance per cost unit is

$$PC(a) = P_a / C_a$$

$$PC(b) = P_b / C_b$$

- the most economical attractive type of computer is the one that maximizes the performance per cost unit, that is

$$\arg \max(\{PC(a), PC(b)\}) = a \text{ ou } b$$



FM2 Pay attention to the scenario described below, and present what is requested.

### **Resolution (cont.) - alternative:**

**c) If  $C_a$  and  $C_b$  are the cost of computer types a and b, present an adequate metric to select the most economically attractive computer.**

- for each type a and b, the average cost per performance unit is

$$CP(a) = C_a / P_a$$

$$CP(b) = C_b / P_b$$

- the most economical attractive type of computer is the one that minimizes the cost per performance unit, that is

$$\arg \min(\{CP(a), CP(b)\}) = a \text{ ou } b$$

# FM2 Pay attention to the scenario described below, and present what is requested.

**Example** (two types of computers,  $a$  and  $b$ ):

- **cost of the computers:**  $C_a = 1000\text{€}$  ;  $C_b = 1500\text{€}$

- **results of the micro-benchmarks:**

$R_x(a) = 10$ ,  $R_y(a) = 7$ ,  $R_z(a) = 34$  ;  $R_x(b) = 1$ ,  $R_y(b) = 77$ ,  $R_z(b) = 43$

- **weighted averages:**

$P_a = 0.2178 * 10 + 0.1089 * 7 + 0.66 * 34 = 25.3803$

$P_b = 0.2178 * 1 + 0.1089 * 77 + 0.66 * 43 = 36.9831$

- **speedups:**

$S(a,b) = P_a / P_b = 0.686$  ;  $S(b,a) = P_b / P_a = 1.457$  ( $a$  is 45.7% faster than  $b$ )

- **performance per cost:**

$PC(a) = P_a / C_a = 25.3803 / 1000 = 0.0253$

$PC(b) = P_b / C_b = 36.9831 / 1500 = 0.0246$

$\arg \max(\{PC(a), PC(b)\}) = a$  ( $a$  maximizes performance per cost)

- **cost per performance:**

$CP(a) = C_a / P_a = 1000 / 25.3803 = 39.4006$

$CP(b) = C_b / P_b = 1500 / 36.9831 = 40.5590$

$\arg \min(\{CP(a), CP(b)\}) = a$  ( $a$  minimizes cost per performance)

**FM3 Pay attention to the scenario described below, and present what is requested.**

**a) On a YouTube channel dedicated to hardware analysis, CPUs X and Y were compared, based on the same computer, to find out which CPU was faster. The comparison consisted of carrying out 5 different benchmarks, b1 to b5, of equal importance (equal weight), obtaining a total of 10 results, Xb1 to Yb5. The results were further normalized considering a reference CPU Z. Present the expression of the geometric mean of the results for CPUs X and Y, compatible with this description.**

**b) Assuming the 5 benchmarks have different weights, w1 to w5, present the respective weighted geometric mean expressions.**

**c) Knowing that for CPU Z the results of the 5 benchmarks are 4, 6, 8, 10, 12 and for CPU X they are 3, 5, 7, 11, 13, present the value of the geometric mean for CPU X, and of the weighted variant (for the weights  $w_1=50\%$ ,  $w_2=25\%$ ,  $w_3=12.5\%$ ,  $w_4=6.25\%$  and  $w_5=6.25\%$ ).**

# FM3 Pay attention to the scenario described below, and present what is requested.

## Resolution a):

- the **normalized results** for CPUs X and Y are:

$$\text{CPU X: } X_{n1} = Z_{b1}/X_{b1}, \dots, X_{n5} = Z_{b5}/X_{b5}$$

$$\text{CPU Y: } Y_{n1} = Z_{b1}/Y_{b1}, \dots, Y_{n5} = Z_{b5}/Y_{b5}$$

- 5 different benchmarks: the geometric mean is a root of index 5
- the **geometric means** for CPUs X and Y are given by:

$$\text{CPU X: } X_g = \text{ROOT5}(X_{n1} * \dots * X_{n5}) = (X_{n1} * \dots * X_{n5})^{(1/5)}$$

$$\text{CPU Y: } Y_g = \text{ROOT5}(Y_{n1} * \dots * Y_{n5}) = (Y_{n1} * \dots * Y_{n5})^{(1/5)}$$

# FM3 Pay attention to the scenario described below, and present what is requested.

## Resolution b):

- the **weighted geometric mean** for CPU X is given by:

$$X_g = X_{n1}^{w1} * X_{n2}^{w2} * X_{n3}^{w3} * X_{n4}^{w4} * X_{n5}^{w5}$$

## Resolution c):

- the **normalized results** for CPU X are:

$$X_{n1}=4/3, X_{n2}=6/5, X_{n3}=8/7, X_{n4}=10/11, X_{n5}=12/13$$

- the **geometric mean** for CPU X is given by:

$$X_g = [ (4/3) * (6/5) * (8/7) * (10/11) * (12/13) ]^{(1/5)} = 1.0894$$

- the **weighted geometric mean** for CPU X is given by:

$$X_g = (4/3)^{(0.5)} * (6/5)^{(0.25)} * (8/7)^{(0.125)} * (10/11)^{(0.0625)} * (12/13)^{(0.0625)} = 1.2155$$

**FM4A Pay attention to the scenario described below, and present what is requested.**



**A game displays on the screen the number of frames per second (FPS) as it unfolds.**

**During a gaming session, the FPS number evolved as follows:  
120 FPS during the first 1000 frames, 60 FPS during the second 1000 frames, and 30 FPS during the third 1000 frames, after which the player gave up continuing to play.**

**Calculate the average FPS number over the time the game lasted.**

# FM4A Pay attention to the scenario described below, and present what is requested.

## Resolution:

- total frames =  $3 * 1000 = 3000$  frames
- time to produce the first 1000 frames:  $1000/120 = 8.33$  s
- time to produce the second 1000 frames:  $1000/60 = 16.66$  s
- time to produce the third 1000 frames:  $1000/30 = 33.33$  s
- total execution time =  $1000/120 + 1000/60 + 1000/30 = 58.33$  s
- **average frame rate =  $3000$  frames /  $58.33$  s =  $51.43$  FPS**
- as the FPS metric is a rate (frames per second), the average performance can also be measured through the harmonic mean

$$3 / (1/120 + 1/60 + 1/30) = 51.43 \text{ FPS}$$

**NOTE:** each portion of frames has the same size (1000) and therefore the non-weighted harmonic mean is used

**FM4B Pay attention to the scenario described below, and present what is requested.**



**A game displays on the screen the number of frames per second (FPS) as it unfolds.**

**During a gaming session, the FPS number evolved as follows:  
120 FPS during the first minute, 60 FPS during the second minute, and 30 FPS during the third minute, after which the player gave up continuing to play.**

**Calculate the average FPS number over the time the game lasted.**



# FM4B Pay attention to the scenario described below, and present what is requested.

## Resolution:

- total execution time =  $60 * 3 = 180$  s
- in the 1st 60s, the total n° of frames was  $60 \text{ s} * 120 \text{ FPS} = 7200$  frames
- in the 2nd 60s, the total n° of frames was  $60 \text{ s} * 60 \text{ FPS} = 3600$  frames
- in the 3rd 60s, the total n° of frames was  $60 \text{ s} * 30 \text{ FPS} = 1800$  frames
- total frames =  $7200 + 3600 + 1800 = 12600$  frames
- **average frame rate** =  $12600 \text{ frames} / 180 \text{ s} = 70 \text{ FPS}$
- the same rate can be obtained by a **weighted harmonic mean**:

$$1 / [(7200/12600)/120 + (3600/12600)/60 + (1800/12600)/30] = 70 \text{ FPS}$$

**NOTE:** each portion of frames has a different size (7200, 3600, 1800) and thus a weighted harmonic mean is used

**FM4C Pay attention to the scenario described below, and present what is requested.**



**A game displays on the screen the number of frames per second (frames per second or FPS) as it unfolds.**

**During a gaming session, the FPS number evolved as follows: 120 FPS during the first minute, 60 FPS during the next three minutes, and 30 FPS during the next two minutes, after which the player gave up continuing to play.**

**Calculate the average FPS number over the time the game lasted.**

# FM4C Pay attention to the scenario described below, and present what is requested.

## Resolution:

- total execution time =  $60 * (1+3+2) = 360$  s
- in the 1st minute, the n° of frames was  $60s * 120 \text{ FPS} = 7200$  frames
- in the next 3 minutes, the n° of frames was  $180s * 60 \text{ FPS} = 10800$  frames
- in the next 2 minutes, the n° of frames was  $120s * 30 \text{ FPS} = 3600$  frames
- total frames =  $7200 + 10800 + 3600 = 21600$  frames
- **average frame rate =  $21600 \text{ frames} / 360 \text{ s} = 60 \text{ FPS}$**
- the same rate can be obtained by a **weighted harmonic mean**:

$$1 / [(7200/21600)/120 + (10800/21600)/60 + (3600/21600)/30] = 60 \text{ FPS}$$

**NOTE:** each portion of frames has a different size (7200, 10800, 3600) and thus a weighted harmonic mean is used

# TE1 Compile the program te1.c (\*) and execute it in the two ways described below. Compare the results and draw conclusions.

(\*) available in the “unit 7” folder, on the Virtual platform

## Compilation in Windows

**- option 1: IDE VSCode and GCC compiler**

(<https://code.visualstudio.com/docs/cpp/config-mingw>)

**- option 2: IDE Geany and GCC compiler**

(instructions in the file *howto-install-and-use-geany-and-gcc-in-windows.txt*, in the “unit 7” folder, on the Virtual platform)

(\*) assuming the terminal was opened in the folder containing the executable te1

### a) Direct execution:

- directly from the VSCode IDE
- (\*) by typing in the terminal “./te1” (in Linux) or “.\te1” (in Windows)

### b) Indirect execution (terminal only) (\*):

- Linux: `time ./te1`

- Windows:

`powershell "Measure-Command{.\te1 | Out-Default}"`

```
#include <stdio.h>
#include <sys/time.h>
int main () {
    // Start measuring time
    struct timeval begin, end;
    gettimeofday(&begin, 0);
    // Task
    int iterations = 1000*1000*1000;
    double sum = 0, add = 1;
    for (int i=0; i<iterations; i++) {
        sum += add; add /= 2.0;
    }
    // Stop measuring time
    gettimeofday(&end, 0);
    // Calculate the elapsed time
    long seconds = end.tv_sec - begin.tv_sec;
    long microseconds = end.tv_usec - begin.tv_usec;
    double elapsed = seconds + microseconds*1e-6;

    printf("Result: %.3f\n", sum);
    printf("Time: %.3f seconds.\n", elapsed);
    return 0;
}
```

**BS1 Install the referenced benchmark and follow the instructions.**

**PassMark PerformanceTest is a cross-platform benchmarking application that evaluates different sub-systems of a computing system, producing performance metrics specific to each sub-system, and a synthetic global metric.**

**a) Download this application from the link (\*), install it, and run it on your computer. At the end, write down the individual and global metrics. Note: before running the application, disable the antivirus and ensure that there are no operating system or application updates in progress.**

**(\*) <https://www.passmark.com/products/performancetest/download.php>;  
see also the folder “unit 7”, in the Virtual platform**

**b) Apply the individual metrics collected in a) to the formula**

$$\text{Passmark Rating} = 1 / (((1 / (\text{CPU Mark} * 0.33)) + (1 / (\text{2D Mark} * 50)) + (1 / (\text{3D Mark} * 0.5)) + (1 / (\text{Memory Mark} * 1.92)) + (1 / (\text{Disk Mark} * 0.37)))) / 5$$

**and check if the value matches the global metric collected in a).**

**Note: this formula is used to generate the global metric, according to**

**<https://forums.passmark.com/performancetest/4599-formula-cpu-mark-memory-mark-and-disk-mark/page2?4490-Formula-CPU-mark-memory-mark-and-disk-mark=>**

TOC1 Compile the program toc1.c (\*) and execute it. Compare the speedups obtained with those shown if the examples of code optimization techniques of section 7.4 of the theoretical slides.

(\*) available in the “unit 7” folder, on the Virtual platform

| optimization technique     | section 7.4 speedup | TOC1 speedup |
|----------------------------|---------------------|--------------|
| loop unrolling             | 1.75                |              |
| loop fusion                | 1.62                |              |
| loop fusion+loop unrolling | 2.14                |              |
| loop peeling               | 1.16                |              |
| data independence          | 1.11                |              |
| loop interchange           | 2.12                |              |
| function call avoidance    | 10.52               |              |

SC1 Compile the program sc1.c (\*) with GCC, using the different options provided bellow. Execute each version 5 times and register the smallest time (\*\*). Compute the speedup relative to version -O0.

(\*) available in the “unit 7” folder, on the Virtual platform

(\*\*) measure the time in the same way as in the indirect execution in exercise TE1

| compilation command                                                 | 5-best execution time(s) | speedup relative to -O0 |
|---------------------------------------------------------------------|--------------------------|-------------------------|
| gcc - <b>O0</b> sc1.c -o sc1_O0.exe -lm                             |                          | 1.0                     |
| gcc - <b>O1</b> sc1.c -o sc1_O1.exe -lm                             |                          |                         |
| gcc - <b>O2</b> sc1.c -o sc1_O2.exe -lm                             |                          |                         |
| gcc - <b>O3</b> sc1.c -o sc1_O3.exe -lm                             |                          |                         |
| gcc - <b>O3 -funroll-loops</b> sc1.c -o sc1_O3_unroll_loops.exe -lm |                          |                         |

LA1 The daytime load of a system consists of 60% CPU activity and 40% disk activity. We may improve the disks performance by 150% for 8000€, and the CPU performance by 40% for 5000€.

- a) Which individual upgrade improves more the overall performance ?
- b) Which individual upgrade offers the best cost/benefit ratio ?
- c) What is the break-even point for the upgrades? That is, what price would we need to charge for the CPU (or the disk – change only one) so the result was the same cost per 1% increase for both?
- d) What is the CPU upgrade performance improvement that is necessary in order to produce the same overall performance improvement as the one produced by the Disk upgrade (and vice-versa) ?
- e) What would be the impact on the overall performance by upgrading the CPU and the disk at the same time ?

*Amdahl's Law:*

$$S = \frac{1}{(1-f) + \frac{f}{k}}$$



LA1 The daytime load of a system consists of 60% CPU activity and 40% disk activity. We may improve the disks performance by 150% for 8000€, and the CPU performance by 40% for 5000€.

**a) Which individual upgrade improves more the overall performance ?**

**SOLUTION:**

*Amdahl's Law:*

$$S = \frac{1}{(1-f) + \frac{f}{k}}$$

$$f_{\text{CPU}} = 60\% = 0.6 ; k_{\text{CPU}} = 100\% + 40\% = 140\% = 1.4$$

$$S_{\text{CPU}} = 1 / ( (1-0.60) + 0.60/1.4 ) = 1.2069 \text{ or } 120.69\% \text{ or } +20.69\%$$

$$f_{\text{DISK}} = 40\% = 0.4 ; k_{\text{DISK}} = 100\% + 150\% = 250\% = 2.5$$

$$S_{\text{DISK}} = 1 / ( (1-0.40) + 0.40/2.5 ) = 1.3158 \text{ or } 131.58\% \text{ or } +31.58\%$$

Thus, the **disk** upgrade yields the best overall performance improvement.

LA1 The daytime load of a system consists of 60% CPU activity and 40% disk activity. We may improve the disks performance by 150% for 8000€, and the CPU performance by 40% for 5000€.

**b) Which individual upgrade offers the best cost/benefit ratio ?**

**SOLUTION:**

$$S_{\text{CPU}} = +20.69\% ; S_{\text{DISK}} = +31.58\% ; \text{Cost}_{\text{CPU}} = 5000\text{€} ; \text{Cost}_{\text{DISK}} = 8000\text{€}$$

$$\text{Cost}(+1\%, \text{CPU}) = \text{Cost}_{\text{CPU}} / S_{\text{CPU}} = 5000\text{€} / +20.69\% = \mathbf{241.66\text{€}}$$

$$\text{Cost}(+1\%, \text{DISK}) = \text{Cost}_{\text{DISK}} / S_{\text{DISK}} = 8000\text{€} / +31.58\% = 253.32\text{€}$$

Thus, the **CPU** upgrade yields the **best cost/benefit ratio**.

LA1 The daytime load of a system consists of 60% CPU activity and 40% disk activity. We may improve the disks performance by 150% for 8000€, and the CPU performance by 40% for 5000€.

**c) What is the break-even point for the upgrades? That is, what price would we need to charge for the CPU (or the disk – change only one) so the result was the same cost per 1% increase for both?**

## **SOLUTION:**

$S_{\text{CPU}} = +20.69\%$  ;  $S_{\text{DISK}} = +31.58\%$ ;  $\text{Cost}_{\text{CPU}} = 5000\text{€}$ ;  $\text{Cost}_{\text{DISK}} = 8000\text{€}$

$\text{Cost}(+1\%, \text{CPU}) = \text{Cost}_{\text{CPU}} / S_{\text{CPU}} = 5000\text{€} / +20.69\% = 241.66\text{€}$

$\text{Cost}(+1\%, \text{DISCO}) = \text{Cost}_{\text{DISK}} / S_{\text{DISK}} = 8000\text{€} / +31.58\% = 253.32\text{€}$

We want the price per 1% to be the same

If X is the new CPU target-cost, then  $X/20.69\% = 253.32\text{€}$ , and **X=5241€**

If Y is the new Disk target-cost, then  $Y/31.58\% = 241.66\text{€}$ , and **Y=7631€**

LA1 The daytime load of a system consists of 60% CPU activity and 40% disk activity. We may improve the disks performance by 150% for 8000€, and the CPU performance by 40% for 5000€.

d) What is the CPU upgrade performance improvement that is necessary in order to produce the same overall performance improvement as the one produced by the Disk upgrade (and vice-versa) ?

## SOLUTION:

$$S = 1 / [ (1-f) + f/k ] \Leftrightarrow \dots \Leftrightarrow k = f / [1/S - (1-f)]$$
$$S_{\text{CPU}} = 1.2069 \text{ or } 120.69\% ; S_{\text{DISK}} = 1.3158 \text{ or } 131.58\%$$

$$k_{\text{CPU}} = f_{\text{CPU}} / [1/S_{\text{DISK}} - (1-f_{\text{CPU}})] = 0.6 / [1/1.3158 - (1 - 0.6)] =$$
$$= 1.6666 ; \text{ that is, the CPU should improve by } \mathbf{66.66\%}$$

$$k_{\text{DISK}} = f_{\text{DISK}} / [1/S_{\text{CPU}} - (1-f_{\text{DISK}})] = 0.4 / [1/1.2069 - (1 - 0.4)] =$$
$$= 1.7500 ; \text{ that is, the Disk should improve by } \mathbf{75.00\%}$$

LA1 The daytime load of a system consists of 60% CPU activity and 40% disk activity. We may improve the disks performance by 150% for 8000€, and the CPU performance by 40% for 5000€.

**e) What would be the impact on the overall performance by upgrading the CPU and the disk at the same time ?**

## **SOLUTION:**

$$F_{\text{CPU}} = 60\% = 0.6 ; k_{\text{CPU}} = 100\% + 40\% = 140\% = 1.4$$

$$F_{\text{DISK}} = 40\% = 0.4 ; k_{\text{DISK}} = 100\% + 150\% = 250\% = 2.5$$

$$S_{\text{CPU+DISK}} = 1 / (0.60/1.4 + 0.4/2.5) = 1.6990 \text{ ou } 169.90\% \text{ or } \mathbf{+69.90\%}$$

Thus, the combined update of CPU and Disk yields a speedup of +69.90% in the global performance of the system.

# LA2 Apply Amdahl's Law in the following scenario.

The load of a system is distributed in 30% by a CPU, 50% by a GPU (graphics card) and 20% by a disk. The system can accommodate an additional CPU as the one installed, an additional GPU as the one installed, and an additional disk as the one installed. An ideal situation is also assumed, in which a pair of identical components translates into a 100% increase in performance compared to an isolated component. Assuming that the cost of a CPU, a GPU and an additional disk is 1000€, 2000€ and 500€, respectively, answer the following questions:

- a) Which individual upgrade improves more the overall performance ?
- b) Which individual upgrade offers the best cost/benefit ratio ?
- c) If buying two types of components together gives a 20% discount on the global cost, what pair of new components offers the best cost/benefit ratio ?

$$\text{Amdahl's Law:}$$
$$S = \frac{1}{(1-f) + \frac{f}{k}}$$

# LA2 Apply Amdahl's Law in the following scenario.

**a) Which individual upgrade improves more the overall performance ?**

**SOLUTION:**

$$f_{\text{CPU}} = 30\% = 0.3 ; f_{\text{GPU}} = 50\% = 0.5 ; f_{\text{Disk}} = 20\% = 0.2$$

$$k_{\text{CPU}} = k_{\text{GPU}} = k_{\text{Disk}} = 200\% = 2.0$$

$$S_{\text{CPU}} = 1 / [ (1 - 0.3) + 0.3/2.0 ] = 1.1764 = +17.64\%$$

$$S_{\text{GPU}} = 1 / [ (1 - 0.5) + 0.5/2.0 ] = 1.33(3) = \mathbf{+33.33\%}$$

$$S_{\text{Disk}} = 1 / [ (1 - 0.2) + 0.2/2.0 ] = 1.11(1) = +11.11\%$$

The best upgrade from an overall performance standpoint is the replacement of the GPU.

## LA2 Apply Amdahl's Law in the following scenario.

**b) Which individual upgrade offers the best cost/benefit ratio ?**

**SOLUTION:**

$$S_{\text{CPU}} = +17.64\% ; S_{\text{GPU}} = +33.33\% ; S_{\text{Disk}} = +11.11\%$$

$$\text{Cost}_{\text{CPU}} = 1000\text{€} ; \text{Cost}_{\text{GPU}} = 2000\text{€} ; \text{Cost}_{\text{Disk}} = 500\text{€}$$

$$\text{Cost}(+1\%, \text{CPU}) = \text{Cost}_{\text{CPU}} / S_{\text{CPU}} = 1000\text{€} / +17.64\% = 56.68 \text{ €}$$

$$\text{Cost}(+1\%, \text{GPU}) = \text{Cost}_{\text{GPU}} / S_{\text{GPU}} = 2000\text{€} / +33.33\% = 60.00 \text{ €}$$

$$\text{Cost}(+1\%, \text{Disk}) = \text{Cost}_{\text{Disk}} / S_{\text{Disk}} = 500\text{€} / +11.11\% = \mathbf{45.00 \text{ €}}$$

The best upgrade from a return on investment standpoint is the replacement of the Disk.



# LA2 Apply Amdahl's Law in the following scenario.

**c) If buying two types of components together gives a 20% discount on the global cost, what pair of new components offers the best cost/benefit ratio ?**

## **SOLUTION:**

$$f_{\text{CPU}} = 0.3 ; f_{\text{GPU}} = 0.5 ; f_{\text{SSD}} = 0.2 ; k_{\text{CPU}} = k_{\text{GPU}} = k_{\text{SSD}} = 2.0$$

Possible pairs: CPU+GPU, CPU+SSD, GPU+SSD

$$\text{Cost}_{\text{CPU+GPU}} = (1000+2000)*0.8 = 2400\text{€}$$

$$\text{Cost}_{\text{CPU+SSD}} = (1000+500)*0.8 = 1200\text{€}$$

$$\text{Cost}_{\text{GPU+SSD}} = (2000+500)*0.8 = 2000\text{€}$$

$$S_{\text{CPU+GPU}} = 1 / ( 0.2 + 0.3/2.0 + 0.5/2.0 ) = 1.666(6) = +66.66\%$$

$$S_{\text{CPU+SSD}} = 1 / ( 0.5 + 0.3/2.0 + 0.2/2.0 ) = 1.333(3) = +33.33\%$$

$$S_{\text{GPU+SSD}} = 1 / ( 0.3 + 0.5/2.0 + 0.2/2.0 ) = 1.5384 = +53.74\%$$

$$\text{Cost}(+1\%, \text{CPU+GPU}) = \text{Cost}_{\text{CPU+GPU}} / S_{\text{CPU+GPU}} = 2400\text{€} / +66.66\% = \mathbf{36,00 \text{ €}}$$

$$\text{Cost}(+1\%, \text{CPU+SSD}) = \text{Cost}_{\text{CPU+SSD}} / S_{\text{CPU+SSD}} = 1200\text{€} / +33.33\% = \mathbf{36.00 \text{ €}}$$

$$\text{Cost}(+1\%, \text{GPU+SSD}) = \text{Cost}_{\text{GPU+SSD}} / S_{\text{GPU+SSD}} = 2000\text{€} / +53.74\% = 37.21 \text{ €}$$

Right choice: update the CPU together with the GPU or the SSD

(choosing one of the alternatives must consider other criteria besides the cost).

Also, the joint upgrade is more sensible than the best individual upgrade (45€/1%).

# LA3 Apply Amdahl's Law in the following scenario.

The load on a system is divided into 40% per CPU and 60% per disk. Consider the following scenarios where the original disk is replaced.

a) Assume that the original disk is a SATA SSD and that it fails, being replaced by a magnetic disk, with a performance of 1/5 of the SSD. What is the impact on overall system performance?

b) Following the breakdown of the original SSD, it was decided to purchase a new unit, with two hypotheses considered: a new SATA SSD with 30% more performance than the old SSD and costing 300€, or a new NVMe SSD which is 4 times faster than the old SSD but costs 500€.

b1) What is the expected improvement in global performance, for each upgrade hypothesis?

b2) What is the best upgrade, from a cost/benefit perspective?

$$\text{Amdahl's Law:}$$
$$S = \frac{1}{(1-f) + \frac{f}{k}}$$

# LA3 Apply Amdahl's Law in the following scenario.

**a) Assume that the original disk is a SATA SSD and that it fails, being replaced by a magnetic disk, with a performance of 1/5 of the SSD. What is the impact on overall system performance?**

**SOLUTION:**

$$f_{\text{CPU}} = 40\% = 0.4 ; f_{\text{Disk}} = 60\% = 0.6$$

$$k_{\text{HD}} = 1/5 = 0.2$$

$$S_{\text{HD}} = 1 / [ (1 - 0.6) + 0.6/0.2 ] = 0.2941 = \mathbf{29.41\% (<==> - 70.59\%)}$$

**Overall performance decreases to 29.41% (i.e. decreases 70.59%).**

# LA3 Apply Amdahl's Law in the following scenario.

**b) Following the breakdown of the original SSD, it was decided to purchase a new unit, with two hypotheses considered: a new SATA SSD with 30% more performance than the old SSD and costing 300€, or a new NVMe SSD which is 4 times faster than the old SSD but costs 500€.**

**b1) What is the expected improvement in global performance, for each upgrade hypothesis?**

**SOLUTION:**

SSD2 = new SATA SSD; SSD3 = new NVMe SSD

$f_{\text{Disk}} = 60\% = 0.6$  ;  $k_{\text{SSD2}} = 130\% = 1.3$  ;  $k_{\text{SSD3}} = 400\% = 4.0$

$S_{\text{SSD2}} = 1 / [ (1 - 0.6) + 0.6/1.3 ] = 1.1607 = \textbf{+16.07\%}$

$S_{\text{SSD3}} = 1 / [ (1 - 0.6) + 0.6/4.0 ] = 1.8181 = \textbf{+81.81\%}$

# LA3 Apply Amdahl's Law in the following scenario.

**b) Following the breakdown of the original SSD, it was decided to purchase a new unit, with two hypotheses considered: a new SATA SSD with 30% more performance than the old SSD and costing 300€, or a new NVMe SSD which is 4 times faster than the old SSD but costs 500€.**

**b2) What is the best upgrade, from a cost/benefit perspective?**

## **SOLUTION:**

SSD2 = new SATA SSD ; SSD3 = new NVMe SSD

$S_{SSD2} = +16.07\%$  ;  $S_{SSD3} = +81.81\%$

$Cost_{SSD2} = 300\text{€}$  ;  $Cost_{SSD3} = 500\text{€}$

$Cost(+1\%, SSD2) = Cost_{SSD2} / S_{SSD2} = 300\text{€} / +16.07\% = 18.66 \text{ €}$

$Cost(+1\%, SSD3) = Cost_{SSD3} / S_{SSD3} = 500\text{€} / +81.81\% = \mathbf{6.11 \text{ €}}$

**The best upgrade from a return on investment point of view is the purchase of SSD3 (NVME SSD).**

# LA4 Apply Amdahl's Law in the following scenario.

Monitoring a file server indicates the following load distribution, in average terms, across its main hardware components: CPU - 20%; RAM - 25%; 1Gbps Network Card - 15%; Hard Drives – 40%. The owner of the system has 1000€ available to invest in replacing or reinforcing components. After analyzing the market, he identified the following options: A - spend 400€ on replacing the CPU with a 20% faster one; B - double the RAM for 500€ (assuming that doubling the RAM capacity will imply doubling its contribution to the system performance); C - spend 300€ on a 10Gbps network card (assuming that the new card will imply a quintuple contribution of the network card to system performance); D - apply 700€ in replacing disks with others that are 50% faster.

Answer the following questions, respecting the indicated precision and conventions.

a) Assuming that it is only possible to update one of the components:

a1) Provide the global speedup (\*), given by Amdahl's Law, provided by each of the individual updates (\*\*) non-percentage value, to the hundredths, truncated).

a2) Provide the cost per 1% (\*) of the global performance increase, for each of the individual upgrades (\*\*) indicate only the integer part of the cost value).

a3) Based on the results of questions a1) and a2), indicate the upgrade option (A, B, C or D) that:

a3.i) most benefits the overall performance;

a3.ii) represents the best option from the point of view of the benefit obtained, for the cost involved.

b) Assuming that several components can be updated together, within the money available to spend (1000€), select the simultaneous updates that: b1) maximize the global speedup; b2) maximize the benefit for the investment made.

c) Before there was a chance to make any updates, the network card failed and was replaced by a 100Mbps card (that will imply a 90% drop in the contribution of the network card to the performance of the system). Using Amdahl's Law, calculate the impact of this failure on the server's global performance (indicate the non-percentage speedup value, to the hundredths, truncated).

$$\text{Amdahl's Law:}$$
$$S = \frac{1}{(1 - f) + \frac{f}{k}}$$

# LA4 Apply Amdahl's Law in the following scenario.

Monitoring a file server indicates the following load distribution, in average terms, across its main hardware components: CPU - 20%; RAM - 25%; 1Gbps Network Card - 15%; Hard Drives - 40%. The owner of the system has 1000€ available to invest in replacing or reinforcing components. After analyzing the market, he identified the following options: A - spend 400€ on replacing the CPU with a 20% faster one; B - double the RAM for 500€ (assuming that doubling the RAM capacity will imply doubling its contribution to the system performance); C - spend 300€ on a 10Gbps network card (assuming that the new card will imply a quintuple contribution of the network card to system performance); D - apply 700€ in replacing disks with others that are 50% faster.

a) Assuming that it is only possible to update one of the components:

a1) Present the global speedup (\*), given by Amdahl's Law, provided by each of the individual updates ((\*) **non-percentage value, to the hundredths, truncated**).

# LA4 Apply Amdahl's Law in the following scenario.

Monitoring a file server indicates the following load distribution, in average terms, across its main hardware components: CPU - 20%; RAM - 25%; 1Gbps Network Card - 15%; Hard Drives - 40%. The owner of the system has 1000€ available to invest in replacing or reinforcing components. After analyzing the market, he identified the following options: A - spend 400€ on replacing the CPU with a 20% faster one; B - double the RAM for 500€ (assuming that doubling the RAM capacity will imply doubling its contribution to the system performance); C - spend 300€ on a 10Gbps network card (assuming that the new card will imply a quintuple contribution of the network card to system performance); D - apply 700€ in replacing disks with others that are 50% faster.

a) Assuming that it is only possible to update one of the components:

a1) Present the global speedup (\*), given by Amdahl's Law, provided by each of the individual updates ((\*) **non-percentage value, to the hundredths, truncated**).

**SOLUTION:**

$$f_{\text{CPU}} = 20\% = 0.2 ; f_{\text{RAM}} = 25\% = 0.25 ; f_{\text{Net}} = 15\% = 0.15 ; f_{\text{Disk}} = 40\% = 0.4$$
$$k_{\text{CPU}} = 1.2 (+20\%); k_{\text{RAM}} = 2.0 (+100\%); k_{\text{Net}} = 5.0 (+400\%); k_{\text{Disk}} = 1.5 (+50\%)$$

$$\text{option A: } S_{\text{CPU}} = 1 / [ (1 - 0.2) + 0.2/1.2 ] = 1.0344827 \dots = \mathbf{1.03 = +3\%}$$

$$\text{option B: } S_{\text{RAM}} = 1 / [ (1 - 0.25) + 0.25/2 ] = 1.142357 \dots = \mathbf{1.14 = 14\%}$$

$$\text{option C: } S_{\text{Net}} = 1 / [ (1 - 0.15) + 0.15/5 ] = 1.136363 \dots = \mathbf{1.13 = +13\%}$$

$$\text{option D: } S_{\text{Disk}} = 1 / [ (1 - 0.4) + 0.4/1.5 ] = 1.153846 \dots = \mathbf{1.15 = +15\%}$$



# LA4 Apply Amdahl's Law in the following scenario.

Monitoring a file server indicates the following load distribution, in average terms, across its main hardware components: CPU - 20%; RAM - 25%; 1Gbps Network Card - 15%; Hard Drives - 40%. The owner of the system has 1000€ available to invest in replacing or reinforcing components. After analyzing the market, he identified the following options: A - spend 400€ on replacing the CPU with a 20% faster one; B - double the RAM for 500€ (assuming that doubling the RAM capacity will imply doubling its contribution to the system performance); C - spend 300€ on a 10Gbps network card (assuming that the new card will imply a quintuple contribution of the network card to system performance); D - apply 700€ in replacing disks with others that are 50% faster.

a) Assuming that it is only possible to update one of the components:

a2) Provide the cost (\*) per 1% of the global performance increase, for each of the individual upgrades ((\* **only the integer part of the cost value**)).

# LA4 Apply Amdahl's Law in the following scenario.

Monitoring a file server indicates the following load distribution, in average terms, across its main hardware components: CPU - 20%; RAM - 25%; 1Gbps Network Card - 15%; Hard Drives - 40%. The owner of the system has 1000€ available to invest in replacing or reinforcing components. After analyzing the market, he identified the following options: A - spend 400€ on replacing the CPU with a 20% faster one; B - double the RAM for 500€ (assuming that doubling the RAM capacity will imply doubling its contribution to the system performance); C - spend 300€ on a 10Gbps network card (assuming that the new card will imply a quintuple contribution of the network card to system performance); D - apply 700€ in replacing disks with others that are 50% faster.

a) Assuming that it is only possible to update one of the components:

a2) Provide the cost (\*) per 1% of the global performance increase, for each of the individual upgrades ((\* **only the integer part of the cost value**)).

**SOLUTION:**

$$S_{\text{CPU}} = 1.03 = +3\% ; S_{\text{RAM}} = 1.14 = +14\% ; S_{\text{Net}} = 1.13 = +13\% ; S_{\text{Disk}} = 1.15 = +15\%$$
$$\text{Cost}_{\text{CPU}} = 400\text{€} ; \text{Cost}_{\text{RAM}} = 500\text{€} ; \text{Cost}_{\text{Net}} = 300\text{€} ; \text{Cost}_{\text{Disk}} = 700\text{€}$$

$$\text{option A: Cost}(+1\%, \text{CPU}) = \text{Cost}_{\text{CPU}} / S_{\text{CPU}} = 400\text{€} / +3\% = 133.3(3) = \mathbf{133 \text{ €}}$$

$$\text{option B: Cost}(+1\%, \text{RAM}) = \text{Cost}_{\text{RAM}} / S_{\text{RAM}} = 500\text{€} / +14\% = 35.71... = \mathbf{35 \text{ €}}$$

$$\text{option C: Cost}(+1\%, \text{Net}) = \text{Cost}_{\text{Net}} / S_{\text{Net}} = 300\text{€} / +13\% = 23.076... = \mathbf{23 \text{ €}}$$

$$\text{option D: Cost}(+1\%, \text{Disk}) = \text{Cost}_{\text{Disk}} / S_{\text{Disk}} = 700\text{€} / +15\% = 46.6(6)... = \mathbf{46 \text{ €}}$$

# LA4 Apply Amdahl's Law in the following scenario.

Monitoring a file server indicates the following load distribution, in average terms, across its main hardware components: CPU - 20%; RAM - 25%; 1Gbps Network Card - 15%; Hard Drives - 40%. The owner of the system has 1000€ available to invest in replacing or reinforcing components. After analyzing the market, he identified the following options: A - spend 400€ on replacing the CPU with a 20% faster one; B - double the RAM for 500€ (assuming that doubling the RAM capacity will imply doubling its contribution to the system performance); C - spend 300€ on a 10Gbps network card (assuming that the new card will imply a quintuple contribution of the network card to system performance); D - apply 700€ in replacing disks with others that are 50% faster.

a) Assuming that it is only possible to update one of the components:

a3) Based on the results of questions a1) and a2), indicate the upgrade option (A, B, C or D) that:

a3.i) most benefits the overall performance;

a.3.ii) represents the best option from the point of view of the benefit obtained, for the cost involved.

# LA4 Apply Amdahl's Law in the following scenario.

Monitoring a file server indicates the following load distribution, in average terms, across its main hardware components: CPU - 20%; RAM - 25%; 1Gbps Network Card - 15%; Hard Drives - 40%. The owner of the system has 1000€ available to invest in replacing or reinforcing components. After analyzing the market, he identified the following options: A - spend 400€ on replacing the CPU with a 20% faster one; B - double the RAM for 500€ (assuming that doubling the RAM capacity will imply doubling its contribution to the system performance); C - spend 300€ on a 10Gbps network card (assuming that the new card will imply a quintuple contribution of the network card to system performance); D - apply 700€ in replacing disks with others that are 50% faster.

a) Assuming that it is only possible to update one of the components:

a3) Based on the results of questions a1) and a2), indicate the upgrade option (A, B, C or D) that:

a3.i) most benefits the overall performance;

a.3.ii) represents the best option from the point of view of the benefit obtained, for the cost involved.

**SOLUTION:**

a.3.i) option **D** (highest speedup)

a.3.ii) option **C** (option that ensures the cheapest cost per +1% of speedup)

# LA4 Apply Amdahl's Law in the following scenario.

b) Assuming that several components can be updated together, within the money available to spend (1000€), select the simultaneous updates that: b1) maximize the global speedup; b2) maximize the benefit for the investment made. **(use values to the thousands, truncated)**

# LA4 Apply Amdahl's Law in the following scenario.

b) Assuming that several components can be updated together, within the money available to spend (1000€), select the simultaneous updates that: b1) maximize the global speedup; b2) maximize the benefit for the investment made. **(use values to the thousands, truncated)**

## SOLUTION:

A = CPU update ; B = RAM update; C = Net update ; D = Disk update

$f_{\text{CPU}} = 0.2$  ;  $f_{\text{RAM}} = 0.25$  ;  $f_{\text{Net}} = 0.15$  ;  $f_{\text{Disk}} = 0.4$

$k_{\text{CPU}} = 1.2$  ;  $k_{\text{RAM}} = 2.0$  ;  $k_{\text{Net}} = 5.0$  ;  $k_{\text{Disk}} = 1.5$

$\text{Cost}_{\text{CPU}} = 400\text{€}$  ;  $\text{Cost}_{\text{RAM}} = 500\text{€}$  ;  $\text{Cost}_{\text{Net}} = 300\text{€}$  ;  $\text{Cost}_{\text{Disk}} = 700\text{€}$

**possible combinations of at least two costs, with total  $\leq 1000\text{€}$  :**

**A+B = 900€; A+C = 700€; B+C = 800€; C+D = 1000€**

b1)

**Global Speedup with 2 simultaneous updates X and Y:**

$$S_{X,Y} = 1 / [ (1-(f_X+f_Y)) + f_X/k_X + f_Y/k_Y ]$$

$$S_{A,B} = 1 / [(1-(0.2+0.25)) + 0.2/1.2 + 0.25/2.0] = 1.188 = +18.8\%$$

$$S_{A,C} = 1 / [(1-(0.2+0.15)) + 0.2/1.2 + 0.15/5.0] = 1.181 = +18.1\%$$

$$S_{B,C} = 1 / [(1-(0.25+0.15)) + 0.25/2.0 + 0.15/5.0] = 1.324 = +32.4\%$$

$$S_{C,D} = 1 / [(1-(0.15+0.4)) + 0.15/5.0 + 0.4/1.5] = \mathbf{1.339 = +33.9\% (C,D)}$$

# LA4 Apply Amdahl's Law in the following scenario.

b) Assuming that several components can be updated together, within the money available to spend (1000€), select the simultaneous updates that: b1) maximize the global speedup; b2) maximize the benefit for the investment made. **(use values to the thousands, truncated)**

## SOLUTION:

A = CPU update ; B = RAM update; C = Net update ; D = Disk update  
Cost<sub>CPU</sub> = 400€ ; Cost<sub>RAM</sub> = 500€ ; Cost<sub>Net</sub> = 300€ ; Cost<sub>Disk</sub> = 700€

**possible combinations of at least two costs, with total ≤ 1000€ :**

**A+B = 900€; A+C = 700€; B+C = 800€; C+D = 1000€**

b1)  $S_{A,B} = 1.188 = +18.8\%$  ;  $S_{A,C} = 1.181 = +18.1\%$  ;  $S_{B,C} = 1.324 = +32.4\%$  ;  $S_{C,D} = 1.339 = +33.9\%$

b2)

**Cost per each additional 1% with 2 simultaneous updates X and Y:**

Cost(+1%, A, B) = (Cost<sub>A</sub>+Cost<sub>B</sub>) /  $S_{A,B}$  = (400+500)€ / +18.8% = 47.872€

Cost(+1%, A, C) = (Cost<sub>A</sub>+Cost<sub>C</sub>) /  $S_{A,C}$  = (400+300)€ / +18.1% = 38.674€

Cost(+1%, B, C) = (Cost<sub>B</sub>+Cost<sub>C</sub>) /  $S_{B,C}$  = (500+300)€ / +32.4% = **24.691€ (B,C)**

Cost(+1%, C, D) = (Cost<sub>C</sub>+Cost<sub>D</sub>) /  $S_{C,D}$  = (300+700)€ / +33.9% = 29.498€

## LA4 Apply Amdahl's Law in the following scenario.

c) Before there was a chance to make any updates, the network card failed and was replaced by a 100Mbps card (that will imply a 90% drop in the contribution of the network card to the performance of the system). Using Amdahl's Law, calculate the impact of this failure on the server's global performance (indicate the non-percentage speedup value, to the hundredths, truncated).



# LA4 Apply Amdahl's Law in the following scenario.

c) Before there was a chance to make any updates, the network card failed and was replaced by a 100Mbps card (that will imply a 90% drop in the contribution of the network card to the performance of the system). Using Amdahl's Law, calculate the impact of this failure on the server's global performance (provide the non-percentage speedup value, to the hundredths, truncated).

**SOLUTION:**

- $f_{\text{Net}} = 15\% = 0.15$
- a drop of 90% in the performance of the network card implies that  $K_{\text{Net}} = 1.0 - 0.9 = 0.1$
- thus,  $S_{\text{Net}} = 1 / [ (1 - 0.15) + 0.15/0.1 ] = 0.42553... = 0.42$